In [1]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("/home/stu/Downloads/Mall_Customers.csv")
df
```

Out[1]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [2]:
```
df.isnull()
```

Out[2]:

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
195	False	False	False	False	False
196	False	False	False	False	False
197	False	False	False	False	False
198	False	False	False	False	False
199	False	False	False	False	False

200 rows × 5 columns

In [4]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("/home/stu/Downloads/data.csv")
df
```

Out[4]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)	Genre
0	1	19	15	39	3
1	2	21	15	81	6
2	3	20	16	6	9
3	4	23	16	77	4
4	5	31	17	40	3
5	6	22	17	76	5
6	7	35	18	6	2
7	8	23	18	94	3
8	9	64	19	3	1
9	10	30	19	72	4

In [5]:
```
df.mean()
```

Out[5]:
```
CustomerID                 5.5
Age                       28.8
Annual Income (k$)        17.0
Spending Score (1-100)    49.4
Genre                      4.0
dtype: float64
```

In [8]:
```
df.loc[:,'Age'].mean()
```

Out[8]:
```
28.8
```

In [7]:
```
df['Age'].mean()
```

Out[7]:
```
28.8
```

In [10]:
```
df.mean(axis=1) [0:4]
```

Out[10]:
```
0    15.4
1    25.0
2    10.8
3    24.8
dtype: float64
```

In [11]:
```
df.median()
```

Out[11]:
```
CustomerID                 5.5
Age                       23.0
Annual Income (k$)        17.0
Spending Score (1-100)    56.0
Genre                      3.5
dtype: float64
```

In [12]:
```
df.loc[:,'Age'].median()
```

Out[12]:
```
23.0
```

In [14]:
```
df.median(axis=1) [0:5]
```

Out[14]:
```
0    15.0
1    15.0
2     9.0
3    16.0
4    17.0
dtype: float64
```

In [15]:
```
df.mode()
```

Out[15]:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)	Genre
0	1	23.0	15.0	6.0	3.0
1	2	NaN	16.0	NaN	NaN
2	3	NaN	17.0	NaN	NaN
3	4	NaN	18.0	NaN	NaN
4	5	NaN	19.0	NaN	NaN
5	6	NaN	NaN	NaN	NaN
6	7	NaN	NaN	NaN	NaN
7	8	NaN	NaN	NaN	NaN
8	9	NaN	NaN	NaN	NaN
9	10	NaN	NaN	NaN	NaN

In [16]:
```
df.mean(axis=1)
```

Out[16]:
```
0    15.4
1    25.0
2    10.8
3    24.8
4    19.2
5    25.2
6    13.6
7    29.2
8    19.2
9    27.0
dtype: float64
```

In [17]:
```
df.loc[:,'Annual Income (k$)'].median()
```

Out[17]:
```
17.0
```

In [18]:
```
df.loc[:,'Annual Income (k$)'].mode()
```

Out[18]:
```
0    15
1    16
2    17
3    18
4    19
Name: Annual Income (k$), dtype: int64
```

In [19]:
```
df.min()
```

Out[19]:
```
CustomerID                 1
Age                       19
Annual Income (k$)        15
Spending Score (1-100)     3
Genre                      1
dtype: int64
```

In [20]:
```
df.loc[:,'Age'].min(skipna=False)
```

Out[20]:
```
19
```

In [21]:
```
df.max()
```

Out[21]:
```
CustomerID                10
Age                       64
Annual Income (k$)        19
Spending Score (1-100)    94
Genre                      9
dtype: int64
```

In [22]:
```
df.loc[:,'Age'].max(skipna=False)
```

Out[22]:
```
64
```

In [23]:
```
df.std()
```

Out[23]:
```
CustomerID                 3.027650
Age                       13.464356
Annual Income (k$)         1.490712
Spending Score (1-100)    35.094159
Genre                      2.260777
dtype: float64
```

In [24]:
```
df.loc[:,'Age'].std(skipna=False)
```

Out[24]:
```
13.464356237447408
```

In [25]:
```
df.std(axis=1) [0:4]
```

Out[25]:
```
0    15.257785
1    32.179186
2     7.049823
3    30.293564
dtype: float64
```

In [26]:
```
df.groupby(['Genre'])['Age'].mean()
```

Out[26]:
```
Genre
1    64.000000
2    35.000000
3    24.333333
4    26.500000
5    22.000000
6    21.000000
9    20.000000
Name: Age, dtype: float64
```

In [27]:
```
df.groupby(['Annual Income (k$)'])['Age'].mean()
```

Out[27]:
```
Annual Income (k$)
15    20.0
16    21.5
17    26.5
18    29.0
19    47.0
Name: Age, dtype: float64
```

In [ ]: